DEEPESH M

Electronics & Communication Engineering | Bengaluru, Karnataka
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OBJECTIVE

Final-year ECE student (CGPA 9.08) with hands-on experience in UVM-based verification, constrained-random testing, RTL-to-GDSII physical design with timing closure (OpenROAD), FPGA implementation and protocol debugging (UART, SPI, I2C). Cadence tools from coursework: Xcelium, Virtuoso, Innovus, NCLaunch. Seeking a VLSI Design Verification / Post-Silicon Validation internship with full-time conversion.

EDUCATION

BE (Electronics & Communication Engineering) | New Horizon College of Engineering, Bengaluru | 2023 - 2027 | CGPA: 9.08
PUC / 12th: Sri Chaitanya Techno School, Hosur | 78.6% | 2023 **SSLC / 10th:** Sri Chaitanya Techno School, Chennai | 82.6% | 2021

TECHNICAL SKILLS

- **Languages & HDL:** C, C++, Python; Verilog, VHDL, SystemVerilog, UVM
- **EDA Tools:** Cadence Virtuoso, Innovus, NCLaunch, Xcelium; ModelSim, Quartus Prime, OpenROAD (ORFS), Yosys, KLayout, MATLAB
- **Hardware & Debug:** UART, SPI, I2C capture, decode and fault isolation; static timing analysis (setup, hold, removal); ESP32, Raspberry Pi Pico, Arduino Nano; KiCad (schematic, layout, custom footprints)
- **OS & Version Control:** Linux, Windows; Git, Docker

PROJECTS

APB UVM Verification Testbench
Personal Project | Tools: SystemVerilog, UVM 1.2, Cadence Xcelium (EDA Playground), Icarus Verilog, GTKWave

- Built a directed APB testbench in SystemVerilog for a 4-register slave, verifying register access and a PREADY-based wait state using Icarus Verilog and GTKWave.
- Refactored it into a full UVM environment (driver, monitor, sequencer, agent, env, test); ran 5 transactions on Cadence Xcelium with 0 UVM_ERROR/UVM_FATAL and confirmed register writes and wait-state timing through waveform analysis.
- Debugged a PREADY polling issue in the driver by looping on the ready status instead of using a fixed transfer length.

USB Logic Analyzer - Raspberry Pi Pico
Personal Project | Tools: RP2040 (PIO/DMA), Python, KiCad, ESP32

- Designed a 6-channel logic analyzer using PIO + DMA sampling on RP2040, with a custom binary capture protocol streamed over USB serial.
- Built Python decoders for UART, SPI and I2C, each validated against real hardware captures (ESP32 sender, MPU6050 sensor).
- Isolated a silent I2C bus fault by separating wiring, capture-timing and library-level causes; traced it to a regression in the ESP32 Arduino core I2C driver rather than the hardware.
- Designed a DRC-clean 2-layer KiCad PCB with 5V-tolerant input protection and I2C pull-ups; full source published at github.com/mdeepesh232-design/usb-logic-analyzer.

Single-Cycle RISC-V Core - RTL-to-GDSII and Verification
Personal Project | Tools: Verilog, Yosys, OpenROAD (ORFS), Nangate45, Docker

- Designed a single-cycle RISC-V core (12 instructions, 32x32 register file, 1,024 flops) and took it through RTL-to-GDSII on Nangate45; closed timing at 400 MHz (2.5 ns) with setup +0.21 ns, hold +0.11 ns and removal +0.12 ns at ~13,800 μm^2.
- Swept the clock from 10 ns to 2 ns, fitted a slack model (slack ≈ 0.6×T - 1.35 ns) and identified reset removal as the next limit (~1.9 ns).
- Debugged a hollow first run (30 flops): a constant instruction made synthesis prune the register file; confirmed via a LOGIC0-driven net, then added memory ports and a top-level wrapper to get a real 1,024-flop baseline.
- Verified with a 22-check directed test, a monitor + scoreboard (135 checks) and constrained-random tests (20 seeds, ~7,900 checks, 0 mismatches); mutation tests confirmed the checkers catch injected bugs. Source: github.com/mdeepesh232-design/riscv-rtl2gds.

NOMA-Based Underwater Acoustic Communication System (FPGA)
Final Year Project | Team of 3 | Tools: MATLAB, Verilog, Quartus Prime, DE1-SoC (Cyclone V)

- Simulated a 2-user power-domain NOMA link over an underwater acoustic channel in MATLAB (BPSK, SIC, Thorp path loss, Rician fading K=3, colored noise).
- Implemented the design as synthesizable Verilog (FSM-based) on a DE1-SoC FPGA with a 50 MHz clock and positive setup slack (+5.3 ns).
- Measured per-user BER vs SNR on the board for both UWA and AWGN channels and compared the results with the MATLAB reference; second conference paper in preparation.

DIY SpO2 Monitor - Custom PCB Design
Personal Project | Tools: KiCad, Arduino Nano, MAX30102, SSD1306 OLED

- Designed the schematic and PCB layout in KiCad with custom footprints for the MAX30102 sensor and SSD1306 OLED; exported JLCPCB-ready Gerbers.

CERTIFICATIONS

- **VSD - Physical Design Flow:** Udemy (Instructor: Kunal Ghosh) - Oct 2026, 6.5 hours
- **Cisco CCNA: Introduction to Networks** - Nov 2025 (Cisco Networking Academy via New Horizon College of Engineering)